

Collection of Problems

*Author: Jaehoon Song**Release: 2025-11-10 21:12:31-05:00***1. Particular Solution Parameter** (10 points)For what value of k , if any, is

$$y = e^{-2x} + ke^{4x}$$

a solution to the differential equation

$$y - \frac{y''}{4} = 5e^{4x}?$$

Answer:

□

2. Tangent Line Interpretation (10 points)An equation for the line tangent to the graph of the differentiable function f at $x = 3$ is

$$y = 4x + 6.$$

Which of the following statements *must* be true?

(I) $f(0) = 6$

(II) $f(3) = 18$

(III) $f'(3) = 4$

Answer:

□

3. General Solution via Separation of Variables (10 points)

What is the general solution to the differential equation

$$\frac{dy}{dx} = \frac{x \cos(x^2)}{4y}, \quad y > 0?$$

Answer:

□

4. Derivative of an Inverse Tangent (5 points)

Compute

$$\frac{d}{dx}(\tan^{-1}(3x)).$$

Answer:

□

5. **Squeeze Theorem Limit** (10 points)

Let

$$g(x) = \sin\left(\frac{\pi}{2}(x+2)\right) + 3, \quad h(x) = -\frac{1}{4}x^3 - \frac{3}{2}x^2 - \frac{9}{4}x + 3.$$

If f is a function satisfying

$$g(x) \leq f(x) \leq h(x) \quad \text{for } -2 < x < 0,$$

what is

$$\lim_{x \rightarrow -1} f(x)?$$

Answer:

□

6. **Implicit Differentiation** (5 points)

If

$$y = \ln(3x + 4y),$$

then compute

$$\frac{dy}{dx}.$$

Answer:

□

7. **Riemann Sum to Integral** (10 points)

Which of the following definite integrals are equal to

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \sin\left(-1 + \frac{5k}{n}\right) \frac{5}{n}?$$

(I) $\int_{-1}^4 \sin x \, dx$

(II) $\int_0^5 \sin(-1 + x) \, dx$

(III) $5 \int_0^1 \sin(-1 + 5x) \, dx$

Answer:

□

8. **Chain Rule Application** (10 points)

If

$$f(x) = (e^{3x} + \sin(2x))^4,$$

then compute

$$f'(x).$$

Answer:

□

9. **Definite Integral Evaluation** (10 points)

Evaluate

$$\int_3^5 \frac{2x^2 + x + 4}{x - 1} dx.$$

Answer:



10. **Product Rule Derivative** (5 points)

Compute

$$\frac{d}{dx}(\cos x \tan x).$$

Answer:



11. **Inverse Function Derivative** (10 points)

For which of the following decreasing functions f does

$$(f^{-1})'(10) = -\frac{1}{8}?$$

- (A) $f(x) = -5x + 15$
- (B) $f(x) = -2x^3 - 2x + 14$
- (C) $f(x) = -x^5 - 4x + 15$
- (D) $f(x) = e^{-2x} - x + 9$

Answer:



12. **Exponential Decay Model** (10 points)

A dose of 400 milligrams of a drug is administered to a patient. The amount of the drug, $A(t)$, in the bloodstream satisfies

$$\frac{dA}{dt} = kA,$$

for constant k . Which of the following could be an expression for $A(t)$?

- (A) $A(t) = 400e^{-0.3t}$
- (B) $A(t) = e^{-0.3t} + 399$
- (C) $A(t) = -3t + 400$

Answer:



13. **Continuity Check** (5 points)

Confirm that the function

$$f(x) = \frac{x^2 - 5x + 4}{x^2 - 6x + 8}$$

is continuous at $x = 4$.

Answer:

□

14. **Instantaneous Rate at a Time** (5 points)

The depth of water t hours after midnight is

$$D(t) = 10 + 4.9 \cos\left(\frac{\pi}{6}t\right).$$

Describe the method to find the instantaneous rate of change (in ft/hr) at 6 A.M.

Answer:

□

15. **Derivative at a Point** (5 points)

Let

$$f(x) = 2x^3 - 4x^2 - 5.$$

What is $f'(-1)$?

Answer:

□

16. **Indefinite Integral** (10 points)

Compute

$$\int \frac{12x^2}{2x + 1} dx.$$

Answer:

□

17. **Limit Evaluation** (5 points)

Evaluate

$$\lim_{x \rightarrow 3} \frac{x - 3}{x^3 - 9x}.$$

Answer:

□

18. **Combination of Derivatives** (5 points)

If $f'(1) = 2$ and $g'(1) = 6$, and

$$h(x) = 5f(x) - 4g(x) + 3x^2 - 2,$$

find $h'(1)$.

Answer:

□

19. **Differential Equation Solutions** (5 points)

What are all solutions to

$$\frac{dy}{dx} = \frac{1}{x \ln 2}?$$

Answer:

□

20. **Related Rates Relationship** (10 points)

Given $U = \frac{k}{N}$ with constant workforce, describe the relationship between $\frac{dU}{dt}$ and $\frac{dN}{dt}$.

Answer:

□

21. **Points of Inflection** (10 points)

Find the x -values where

$$y = e^{-x} + 2xe^{-x} + x^2e^{-x}$$

has inflection points.

Answer:

□

22. **Vertical Tangents** (10 points)

Describe the y -coordinates of points on the curve

$$e^x = \sin y$$

where the tangent line is vertical.

Answer:

□

23. **Inverse Function Derivative** (10 points)

Let f and g be inverse functions that are differentiable for all x . If $f(-5) = 7$ and $g'(7) = 3$, which of the following statements must be false?

(I) $f'(3) = -\frac{1}{3}$

(II) $f'(-5) = \frac{1}{3}$

(III) $f'(7) = \frac{1}{3}$

Answer:

□

24. **Marginal Cost Estimate** (10 points)

The function C gives the production cost for a bakery to produce cakes of a certain type, where $C(x)$ is the cost, in dollars, to produce x of the cakes. The function M defined by

$$M(x) = C(x+1) - C(x)$$

gives the marginal cost, in dollars, to produce cake number $x + 1$. Which of the following gives the best estimate for the marginal cost, in dollars, to produce the 40th cake?

Answer:

□

25. Left Riemann Sum Approximation (10 points)

Which of the following is a left Riemann sum approximation of

$$\int_1^7 (4 \ln x + 2) dx$$

with n subintervals of equal length?

- (A) $\sum_{k=1}^n \left(4 \ln \left(1 + \frac{k-1}{n} \right) + 2 \right) \frac{1}{n}$
- (B) $\sum_{k=1}^n \left(4 \ln \left(\frac{6k}{n} \right) + 2 \right) \frac{6}{n}$
- (C) $\sum_{k=1}^n \left(4 \ln \left(1 + \frac{6(k-1)}{n} \right) + 2 \right) \frac{6}{n}$
- (D) $\sum_{k=1}^n \left(4 \ln \left(1 + \frac{6k}{n} \right) + 2 \right) \frac{6}{n}$

Answer:

□

26. Instantaneous Rate of Change of the Derivative (10 points)

Let $f(x) = -\frac{32}{x} + x^3$ on the closed interval $[1, 4]$. What is the instantaneous rate of change of $f'(x)$ at $x = 2$?

Answer:

□

27. Arc Length (10 points)

What is the length of the curve

$$y = 1 - \cos x$$

from $x = 0$ to $x = 4\pi$?

Answer:

□

28. Squeeze Theorem Limit (10 points)

Let $g(x) = -x^2 - 2x + 3$ and $h(x) = \frac{1}{2}x^2 + x + \frac{13}{2}$. If f is a function satisfying

$$g(x) \leq f(x) \leq h(x) \quad \text{for all } x,$$

what is

$$\lim_{x \rightarrow -1} f(x) ?$$

Answer:

□

29. **Relative Extrema of an Exponential Polynomial** (10 points)

The function f is defined by

$$f(x) = e^{-x}(x^2 + 2x).$$

At what values of x does f have a relative maximum?

Answer:



30. **Value from a Given Derivative** (10 points)

Let f be a differentiable function such that $f(0) = 5.420$ and

$$f'(x) = \sqrt{\sin^2 x + x}.$$

What is the value of $f(2\pi)$?

Answer:



31. **Limit of a Piecewise Function** (10 points)

Define

$$f(x) = \begin{cases} \frac{(x-1)^2(x+1)}{|x-1|}, & x \neq 1, \\ 2, & x = 1. \end{cases}$$

Find

$$\lim_{x \rightarrow 1} f(x).$$

Answer:



32. **Indefinite Integrals** (10 points)

Evaluate each of the following indefinite integrals.

(a) $\int (6x^5 - 18x^2 + 7) dx$

(b) $\int 6x^5 dx - 18x^2 + 7$

Answer:



33. **Indefinite Integrals** (10 points)

Evaluate each of the following indefinite integrals.

(a) $\int (40x^3 + 12x^2 - 9x + 14) dx$

(b) $\int (40x^3 + 12x^2 - 9x) dx + 14$

(c) $\int (40x^3 + 12x^2) dx - 9x + 14$

Answer:



34. **Indefinite Integral** (5 points)

Evaluate the indefinite integral.

$$\int (12t^7 - t^2 - t + 3) dt$$

Answer:



35. **Indefinite Integral** (5 points)

Evaluate the indefinite integral.

$$\int (10w^4 + 9w^3 + 7w) dw$$

Answer:



36. **Indefinite Integral** (5 points)

Evaluate the indefinite integral.

$$\int (z^6 + 4z^4 - z^2) dz$$

Answer:



37. **Function from Derivative** (10 points)

Determine $f(x)$ given that

$$f'(x) = 6x^8 - 20x^4 + x^2 + 9.$$

Answer:



38. **Function from Derivative** (10 points)

Determine $h(t)$ given that

$$h'(t) = t^4 - t^3 + t^2 + t - 1.$$

Answer:



39. Indefinite Integrals (105 points)

For problems 1–21, evaluate the given integral.

(a) $\int (4x^6 - 2x^3 + 7x - 4) dx$

(b) $\int (z^7 - 48z^{11} - 5z^{16}) dz$

(c) $\int (10t^{-3} + 12t^{-9} + 4t^3) dt$

(d) $\int (w^{-2} + 10w^{-5} - 8) dw$

(e) $\int 12 dy$

(f) $\int (\sqrt[3]{w} + 10\sqrt[5]{w^3}) dw$

(g) $\int (\sqrt{x^7} - 7\sqrt[6]{x^5} + 17\sqrt[3]{x^{10}}) dx$

(h) $\int \left(\frac{4}{x^2} + 2 - \frac{1}{8x^3} \right) dx$

(i) $\int \left(\frac{7}{3y^6} + \frac{1}{y^{10}} - \frac{2}{\sqrt[3]{y^4}} \right) dy$

(j) $\int (t^2 - 1)(4 + 3t) dt$

(k) $\int \sqrt{z} \left(z^2 - \frac{1}{4z} \right) dz$

(l) $\int \frac{z^8 - 6z^5 + 4z^3 - 2}{z^4} dz$

(m) $\int \frac{x^4 - \sqrt[3]{x}}{6\sqrt{x}} dx$

(n) $\int (\sin x + 10 \csc^2 x) dx$

(o) $\int (2 \cos w - \sec w \tan w) dw$

(p) $\int (12 + \csc \theta [\sin \theta + \csc \theta]) d\theta$

(q) $\int \left(4e^z + 15 - \frac{1}{6z} \right) dz$

(r) $\int \left(t^3 - \frac{e^{-t} - 4}{e^{-t}} \right) dt$

(s) $\int \left(\frac{6}{w^3} - \frac{2}{w} \right) dw$

(t) $\int \left(\frac{1}{1+x^2} + \frac{12}{\sqrt{1-x^2}} \right) dx$

(u) $\int \left(6 \cos z + \frac{4}{\sqrt{1-z^2}} \right) dz$

40. **Substitution Method Integrals** (80 points)

For problems 1–16, evaluate the given integral using the substitution method.

(a) $\int (8x - 12)(4x^2 - 12x)^4 dx$

(b) $\int 3t^{-4}(2 + 4t^{-3})^{-7} dt$

(c) $\int (3 - 4w)(4w^2 - 6w + 7)^{10} dw$

(d) $\int 5(z - 4)\sqrt[3]{z^2 - 8z} dz$

(e) $\int 90x^2 \sin(2 + 6x^3) dx$

(f) $\int \sec(1 - z) \tan(1 - z) dz$

(g) $\int (15t^{-2} - 5t) \cos(6t^{-1} + t^2) dt$

(h) $\int (7y - 2y^3)e^{y^4 - 7y^2} dy$

(i) $\int \frac{4w + 3}{4w^2 + 6w - 1} dw$

(j) $\int (\cos(3t) - t^2)(\sin(3t) - t^3)^5 dt$

(k) $\int 4\left(\frac{1}{z} - e^{-z}\right) \cos(e^{-z} + \ln z) dz$

(l) $\int \sec^2 v \cdot e^{1 + \tan v} dv$

(m) $\int 10 \sin(2x) \cos(2x) \sqrt{\cos^2(2x) + 5} dx$

(n) $\int \frac{\csc x \cot x}{2 - \csc x} dx$

(o) $\int \frac{6}{7 + y^2} dy$

(p) $\int \frac{1}{\sqrt{4 - 9w^2}} dw$

41. **Function from Derivative with Initial Condition** (10 points)

Determine $f(x)$ given that

$$f'(x) = 12x^2 - 4x$$

and $f(-3) = 17$.

Answer:

□

42. **Function from Derivative with Initial Condition** (10 points)

Determine $g(z)$ given that

$$g'(z) = 3z^3 + \frac{7}{2\sqrt{z}} - e^z$$

and $g(1) = 15 - e$.

Answer:

□

43. Function from Second Derivative with Initial Conditions (10 points)

Determine $h(t)$ given that

$$h''(t) = 24t^2 - 48t + 2,$$

$$h(1) = -9, \text{ and } h(-2) = -4.$$

Answer:

□

44. Substitution Method Integrals (15 points)

Evaluate each of the following integrals.

(a) $\int \frac{3x}{1 + 9x^2} dx$

(b) $\int \frac{3x}{(1 + 9x^2)^4} dx$

(c) $\int \frac{3}{1 + 9x^2} dx$

Answer:

□

45. **Substitution Method Integrals** (65 points)

For problems 1–13, evaluate the given integral using the substitution method.

- (a) $\int (4\sqrt{5+9t} + 12(5+9t)^7) dt$
- (b) $\int (7x^3 \cos(2+x^4) - 8x^3 e^{2+x^4}) dx$
- (c) $\int \left(\frac{6e^{7w}}{(1-8e^{7w})^3} + \frac{14e^{7w}}{1-8e^{7w}} \right) dw$
- (d) $\int (x^4 - 7x^5 \cos(2x^6 + 3)) dx$
- (e) $\int \left(e^z + \frac{4 \sin(8z)}{1+9 \cos(8z)} \right) dz$
- (f) $\int (20e^{2-8w} \sqrt{1+e^{2-8w}} + 7w^3 - 6\sqrt[3]{w}) dw$
- (g) $\int ((4+7t)^3 - 9t \sqrt[4]{5t^2+3}) dt$
- (h) $\int \left(\frac{6x-x^2}{x^3-9x^2+8} - \csc^2\left(\frac{3x}{2}\right) \right) dx$
- (i) $\int (7(3y+2)(4y+3y^2)^3 + \sin(3+8y)) dy$
- (j) $\int \sec^2(2t)[9+7 \tan(2t) - \tan^2(2t)] dt$
- (k) $\int \frac{8-w}{4w^2+9} dw$
- (l) $\int \frac{7x+2}{\sqrt{1-25x^2}} dx$
- (m) $\int z^7(8+3z^4)^8 dz$

Answer:

□

46. **Area Estimation Using Riemann Sums** (45 points)

For problems 1–3, estimate the area of the region between the function and the x -axis on the given interval using $n = 6$ and using

- (a) the right end points of the subintervals for the height of the rectangles,
 - (b) the left end points of the subintervals for the height of the rectangles, and
 - (c) the midpoints of the subintervals for the height of the rectangles.
- (a) $f(x) = x^3 - 2x^2 + 4$ on $[1, 4]$
 - (b) $g(x) = 4 - \sqrt{x^2 + 2}$ on $[-1, 3]$
 - (c) $h(x) = -x \cos\left(\frac{x}{3}\right)$ on $[0, 3]$

Answer:

□

47. **Net Area Estimation** (5 points)

Estimate the net area between $f(x) = 8x^2 - x^5 - 12$ and the x -axis on $[-2, 2]$ using $n = 8$ and the midpoints of the subintervals for the height of the rectangles. Without looking at a graph of the function on the interval, does it appear that more of the area is above or below the x -axis?

Answer:

□

48. **Definite Integral by Definition** (15 points)

For problems 1–2, use the definition of the definite integral to evaluate the integral. Use the right end point of each interval for x_i^* .

(a) $\int_1^4 (2x + 3) dx$

(b) $\int_0^1 6x(x - 1) dx$

Answer:

□

49. **Definite Integral Evaluation** (5 points)

Evaluate:

$$\int_4^4 \frac{\cos(e^{3x} + x^2)}{x^4 + 1} dx$$

Answer:

□

50. **Properties of Definite Integrals** (10 points)

For problems 1–2, determine the value of the given integral given that $\int_6^{11} f(x) dx = -7$ and $\int_6^{11} g(x) dx = 24$.

(a) $\int_{11}^6 9f(x) dx$

(b) $\int_6^{11} (6g(x) - 10f(x)) dx$

Answer:

□

51. **Properties of Definite Integrals** (5 points)

Determine the value of $\int_2^9 f(x) dx$ given that $\int_5^2 f(x) dx = 3$ and $\int_5^9 f(x) dx = 8$.

Answer:

□

52. **Properties of Definite Integrals** (5 points)

Determine the value of $\int_{-4}^{20} f(x) dx$ given that $\int_{-4}^0 f(x) dx = -2$, $\int_{31}^0 f(x) dx = 19$, and $\int_{20}^{31} f(x) dx = -21$.

Answer:

□

53. Area Interpretation of Definite Integrals (10 points)

For problems 1–2, sketch the graph of the integrand and use the area interpretation of the definite integral to determine the value of the integral.

(a) $\int_1^4 (3x - 2) dx$

(b) $\int_0^5 (-4x) dx$

Answer:

□

54. Fundamental Theorem of Calculus (15 points)

For problems 1–3, differentiate each of the following integrals with respect to x .

(a) $\int_4^x 9 \cos^2(t^2 - 6t + 1) dt$

(b) $\int_7^{\sin(6x)} \sqrt{t^2 + 4} dt$

(c) $\int_{3x^2}^{-1} \frac{e^t - 1}{t} dt$

Answer:

□

55. Definite Integrals (15 points)

Evaluate each of the following integrals.

(a) $\int \left(\cos x - \frac{3}{x^5} \right) dx$

(b) $\int_{-3}^4 \left(\cos x - \frac{3}{x^5} \right) dx$

(c) $\int_1^4 \left(\cos x - \frac{3}{x^5} \right) dx$

Answer:

□

56. **Definite Integrals** (85 points)

Evaluate each of the following integrals, if possible. If it is not possible, clearly explain why it is not possible to evaluate the integral.

(a) $\int_1^6 (12x^3 - 9x^2 + 2) dx$

(b) $\int_{-2}^1 (5z^2 - 7z + 3) dz$

(c) $\int_3^0 (15w^4 - 13w^2 + w) dw$

(d) $\int_1^4 \left(\frac{8}{\sqrt{t}} - 12\sqrt{t^3} \right) dt$

(e) $\int_1^2 \left(\frac{1}{7z} + \frac{\sqrt[3]{z^2}}{4} - \frac{1}{2z^3} \right) dz$

(f) $\int_{-2}^4 \left(x^6 - x^4 + \frac{1}{x^2} \right) dx$

(g) $\int_{-4}^{-1} x^2(3 - 4x) dx$

(h) $\int_2^1 \frac{2y^3 - 6y^2}{y^2} dy$

(i) $\int_0^{\frac{\pi}{2}} (7 \sin t - 2 \cos t) dt$

(j) $\int_0^{\pi} (\sec z \tan z - 1) dz$

(k) $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (2 \sec^2 w - 8 \csc w \cot w) dw$

(l) $\int_0^2 \left(e^x + \frac{1}{x^2 + 1} \right) dx$

(m) $\int_{-5}^{-2} \left(7e^y + \frac{2}{y} \right) dy$

(n) $\int_0^4 f(t) dt$ where

$$f(t) = \begin{cases} 2t & t > 1, \\ 1 - 3t^2 & t \leq 1. \end{cases}$$

(o) $\int_{-6}^1 g(z) dz$ where

$$g(z) = \begin{cases} 2 - z & z > -2, \\ 4e^z & z \leq -2. \end{cases}$$

(p) $\int_3^6 |2x - 10| dx$

(q) $\int_{-1}^0 |4w + 3| dw$

57. **Definite Integrals Using Substitution** (50 points)

Evaluate each of the following integrals, if possible. If it is not possible, clearly explain why it is not possible to evaluate the integral.

(a) $\int_0^1 3(4x + x^4)(10x^2 + x^5 - 2)^6 dx$

(b) $\int_0^{\frac{\pi}{4}} \frac{8 \cos(2t)}{\sqrt{9 - 5 \sin(2t)}} dt$

(c) $\int_{\pi}^0 \sin z \cos^3 z dz$

(d) $\int_1^4 \sqrt{w} e^{1-\sqrt{w^3}} dw$

(e) $\int_{-4}^{-1} \left(\sqrt[3]{5 - 2y} + \frac{7}{5 - 2y} \right) dy$

(f) $\int_{-1}^2 \left(x^3 + e^{\frac{1}{4}x} \right) dx$

(g) $\int_{\pi}^{\frac{3\pi}{2}} (6 \sin(2w) - 7 \cos w) dw$

(h) $\int_1^5 \left(\frac{2x^3 + x}{x^4 + x^2 + 1} - \frac{x}{x^2 - 4} \right) dx$

(i) $\int_{-2}^0 \left(t\sqrt{3 + t^2} + \frac{3}{(6t - 1)^2} \right) dt$

(j) $\int_{-2}^1 ((2 - z)^3 + \sin(\pi z)[3 + 2 \cos(\pi z)]^3) dz$

Answer:

□